7.

| Content | Reactant | | Products | |
|---|---|---|---|---|
| | x | y | *l* | m |
| Stoichiometric coefficient | 2 | 3 | 4 | 1 |
| No. of moles allowed to react | 8 | 15 | – | – |
| No. of moles of reactant reacted and product formed | 8 | 12 | 16 | 4 |
| No. of moles of un-reacted reactants and the product formed | – | 3 | 16 | 4 |

**Limiting reagent** **: x**

**Product formed** **: 16 moles of *l* & 4 moles of m**

**Amount of excess reactant** **: 3 moles of y**

8.

$$\overset{+3}{As_2}\ \overset{-2}{S_3} + H\overset{+5}{N}O_3 \rightarrow H_3\overset{+5}{As}O_4 + H_2\overset{+6}{S}O_4 + \overset{+2}{N}O$$

$2\times2e^-$ $3\times8e^-$ $3e^-$

$4e^- + 24e^-$

$28e^-$

Equate the total no. of electrons in the reactant side by cross multiplying,

$\Rightarrow$ $3\ As_2S_3 + 28\ HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$

Based on reactant side, balance the products

$\Rightarrow$ $3\ As_2S_3 + 28\ HNO_3 \rightarrow 6\ H_3AsO_4 + 9\ H_2SO_4 + 28\ NO$

**Product side** :36 hydrogen atoms & 88 oxygen atoms

**Reactant side** :28 hydrogen atoms & 84 oxygen atoms

Difference is 8 hydrogen atoms & 4 oxygen atoms

$\therefore$ Add 4 $H_2O$ molecule on the reactant side.

Balanced equation is,

**$3\ As_2S_3 + 28\ HNO_3 + 4\ H_2O \rightarrow 6\ H_3AsO_4 + 9\ H_2SO_4 + 28\ NO$**



11th Chemistry Appendix Vol-1 EM.indd 238 11-01-2023 10:54:34